Probability

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Conditional probability

- Conditional probability in statistics measures the probability that a certain event will occur based on the occurrence (or non-occurrence) of other, related events. It has wide applications in science and finance.
- Formally, it is known as:

$$P(\, A \mid B \,) = \frac{P(A \cap B)}{P(B)}$$

- Where:
- $P(A \mid B)$ is the probability that an event A happens given B.
- $P(A \cap B)$ is the probabilty that A and B happens at the same time.
- $P(B)$ is the probability that B happens (must be non negative).

Example of conditional probability

- Imagine we have a deck of 52 cards.
  - A = “the card is an ace”
  - B = “the card is red”
- Now, we know that there are 26 red cards(P(B)), and only 2 red aces ($P(A \cap B)$).
- Therefor, the probability of both happening at the same time is the following:

$$P(A \mid B) = \frac{2}{26} = \frac{1}{13}$$

Bayes' Theorem

- Bayes' theorem is used to calculate conditional probabilities when dealing with uncertain events. In investing, this allows you to update your probability estimates of a market outcome when you get new relevant data.
- Another example is if you want to know the probability that it will rain in the afternoon, it helps to know if it was cloudy in the morning. Conversely, if you know that it did *not* rain in the afternoon, you could calculate the likelihood that it was sunny in the morning.

Bayes' Theorem

- The Bayes' theorem is expressed like this:

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

- Where:
  - $P(A)$ is the previous probability (prior): The initial probability of event (A) before seeing the evidence.
  - $P(B \mid A)$ is the likelihood: The probability of evidence (B) occurring given that hypothesis (A) is true.
  - $P(B)$ is the marginal likelihood: The total probability of the evidence occurring.
  - $P(A \mid B)$ is the posterior probability: The likelihood of event (A) occurring given that (B) is true.

Law of total probability

- The law of total probability is a fundamental principle in probability theory that calculates the overall probability of an event (B) by summing its conditional probabilities across all possible, mutually exclusive, and exhaustive scenarios (Ai). It breaks down complex, multi-stage problems by weighting the probability of an outcome in each scenario by the probability of that scenario occurring.
- The probability P(B) can be calculated considering every possible event that could cause that B happens. If $A1$, $A2$, … , An are mutually exclusive and exhaustive events (i.e., they cover all possible causes of B), then:

$$P(B) = \sum_i P(B|A_i) \cdot P(A_i)$$

- This expression is the total probability of B.

Example of Bayes' Theorem

- Let's suppose that there's a rare illness that affects 1% of the population. P(sick) = 0.01.
- There's a test that gives a true positive 99% of the time when someone has that illness. P(positive|sick) = 0.99.
- but there's also false positives on 5% of healthy people. P(positive|not sick) = 0.05.
- Now, we want to calculate the probability that a person has the illness, given that the test gave positive. P(sick|positive)
- In this instance A = sick, B = positive.
- Applying Bayes' Theorem we have:

$$P(sick|positive) = \frac{P(positive|sick) \cdot P(sick)}{P(positive)}$$

Example of Bayes' Theorem

- To calculate P(B) or P(positive) we have the following equation:
  - $P(positive) = P(positive|sick) \cdot P(sick) + P(positive|\ not\ sick) \cdot P(not\ sick)$
- Replacing values:

$$P(positivo) = (0.99 \cdot 0.01) + (0.05 \cdot 0.99) = 0.0594$$

- Finally:

$$P(enfermedad|positivo) = \frac{(0.99) \cdot (0.01)}{0.0594} = 0.1667$$

References

- https://www.investopedia.com/terms/c/conditional_probability.asp